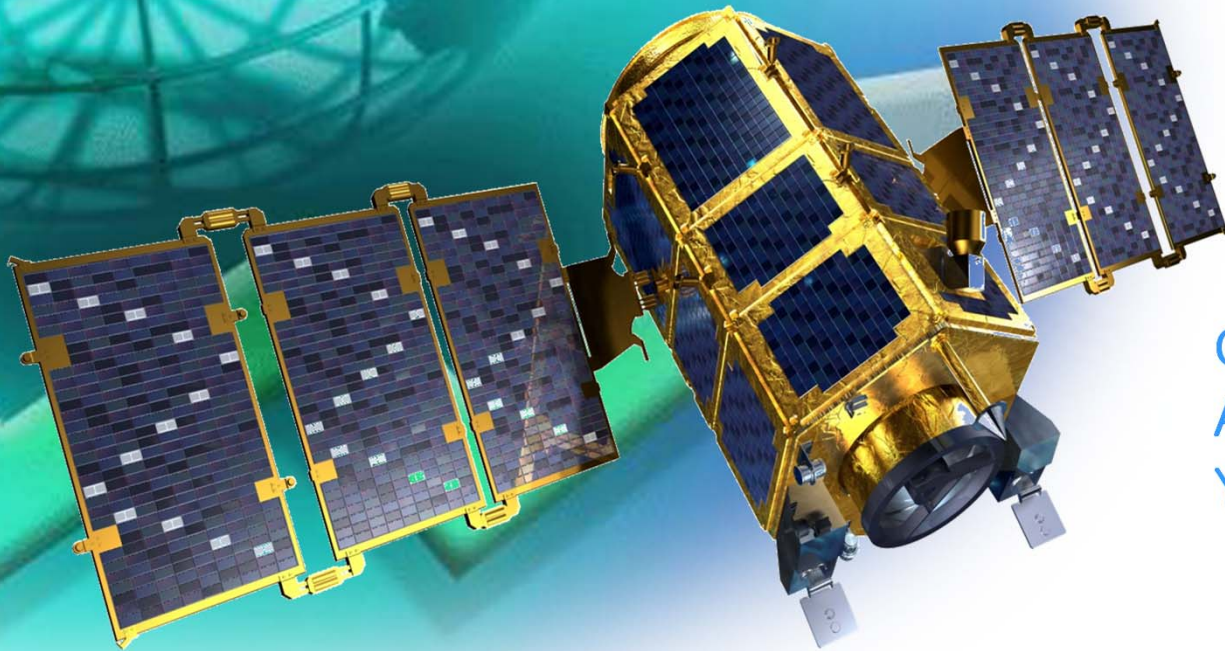
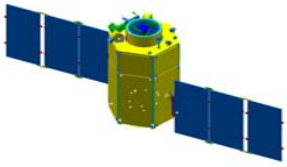


Fall 2011

Spacecraft Systems



Chandeok Park
Astrodynamics & Control Lab.
Yonsei University



Chapter 17

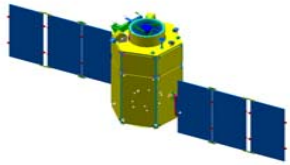
SPACE PROPULSION SYSTEMS

17.1 Propulsion Subsystem Selection and Sizing

17.2 Basics of Rocket Propulsion

17.3 Types of Rockets

17.4 Component Selection and Sizing

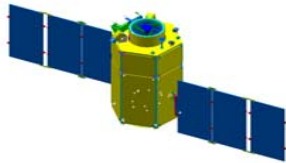


◆ Space Propulsion Systems do three things:

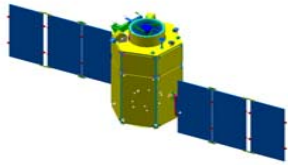
- Lift launch vehicle and its payload (S/C) from the launch pad and place the payload (S/C) into low-earth orbit
- Transfer spacecraft from low-earth orbits into higher orbits or interplanetary trajectories
- Provide thrust for attitude and orbit corrections

TABLE 17-1. Typical Functions and Requirements for Space Propulsion. The change in velocity required for orbit maneuvers is called ΔV . See Table 7-3 in Sec. 7.3 for specific ΔV requirements.

Propulsion Function	Typical Requirement
<i>Orbit transfer to GEO (orbit insertion)</i> • Perigee burn • Apogee burn	2,400 m/s 1,500 (low inclination) to 1,800 m/s (high inclination)
<i>Initial spinup</i>	1 to 60 rpm
<i>LEO to higher orbit raising ΔV</i> • Drag-makeup ΔV • Controlled-reentry ΔV	60 to 1,500 m/s 60 to 500 m/s 120 to 150 m/s



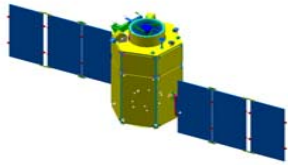
<i>Acceleration to escape velocity from LEO parking orbit</i>	3,600 to 4,000 m/s into planetary trajectory
<i>On-orbit operations (orbit maintenance)</i> • Despin • Spin control • Orbit correction ΔV • East-West stationkeeping ΔV • North-South stationkeeping ΔV • Survivability or evasive maneuvers (highly variable) ΔV	60 to 0 rpm ± 1 to ± 5 rpm 15 to 75 m/s per year 3 to 6 m/s per year 45 to 55 m/s per year 150 to 4,600 m/s
<i>Attitude control</i> • Acquisition of Sun, Earth, Star • On-orbit normal mode control with 3-axis stabilization, limit cycle • Precession control (spinners only) • Momentum management (wheel unloading) • 3-axis control during ΔV	3–10% of total propellant mass Low total impulse, typically <5,000 N*s, 1 K to 10 K pulses, 0.01 to 5.0 sec pulse width 100 K to 200 K pulses, minimum impulse bit of 0.01 N*s, 0.01 to 0.25 sec pulse width Low total impulse, typically <7,000 N*s, 1 K to 10 K pulses, 0.02 to 0.20 sec pulse width 5 to 10 pulse trains every few days, 0.02 to 0.10 sec pulse width On/off pulsing, 10 K to 100 K pulses, 0.05 to 0.20 sec pulse width



17.1 Propulsion Subsystem Selection & Sizing

TABLE 17-2. Propulsion Subsystem Selection and Sizing Process.

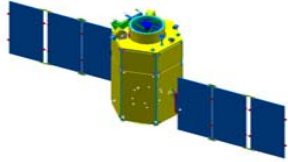
Step	Description of Process	Reference
1	List applicable spacecraft propulsion functions, e.g., orbit insertion, orbit maintenance, attitude control, and controlled de-orbit or reentry	Table 17-1
2	Determine ΔV budget and thrust level constraints for orbit insertion and maintenance	Sec. 7.3
3	Determine total impulse for attitude control, thrust levels for control authority, duty cycles (% on/off, total number of cycles) and mission life requirements	Table 17-1, Sec. 10.3, Chap. 11
4	Determine propulsion system options: • Combined or separate propulsion systems for orbit and attitude control • High vs. low thrust • Liquid vs. solid vs. electric propulsion technology	Secs. 17.1, 17.3
5	Estimate key parameters for each option • Effective I_{sp} for orbit and attitude control • Propellant mass • Propellant and pressurant volume • Configure the subsystem and create equipment list	Secs. 17.1, 17.3 Table 17-3 Tables 17-4, 17-6, 17-7
6	Estimate total mass and power for each option	Table 10-7
7	Establish baseline propulsion subsystem	
8	Document results and iterate as required	



17.1 PS Selection & Sizing

TABLE 17-3. Principal Options for Spacecraft Propulsion Subsystems. See Sec. 17.3 for a definition of specific rocket types.

Propulsion Technology	Orbit Insertion		Orbit Maintenance and Maneuvering	Attitude Control	Typical Steady State I_{sp} (s)
	Perigee	Apogee			
<i>Cold Gas</i>			✓	✓	30–70
<i>Solid</i>	✓	✓			280–300
<i>Liquid</i>					
Monopropellant			✓	✓	220–240
Bipropellant	✓	✓	✓	✓	305–310
Dual mode	✓	✓	✓	✓	313–322
Hybrid	✓	✓	✓		250–340
<i>Electric</i>		✓	✓		300–3,000



17.1 PS Selection & Sizing

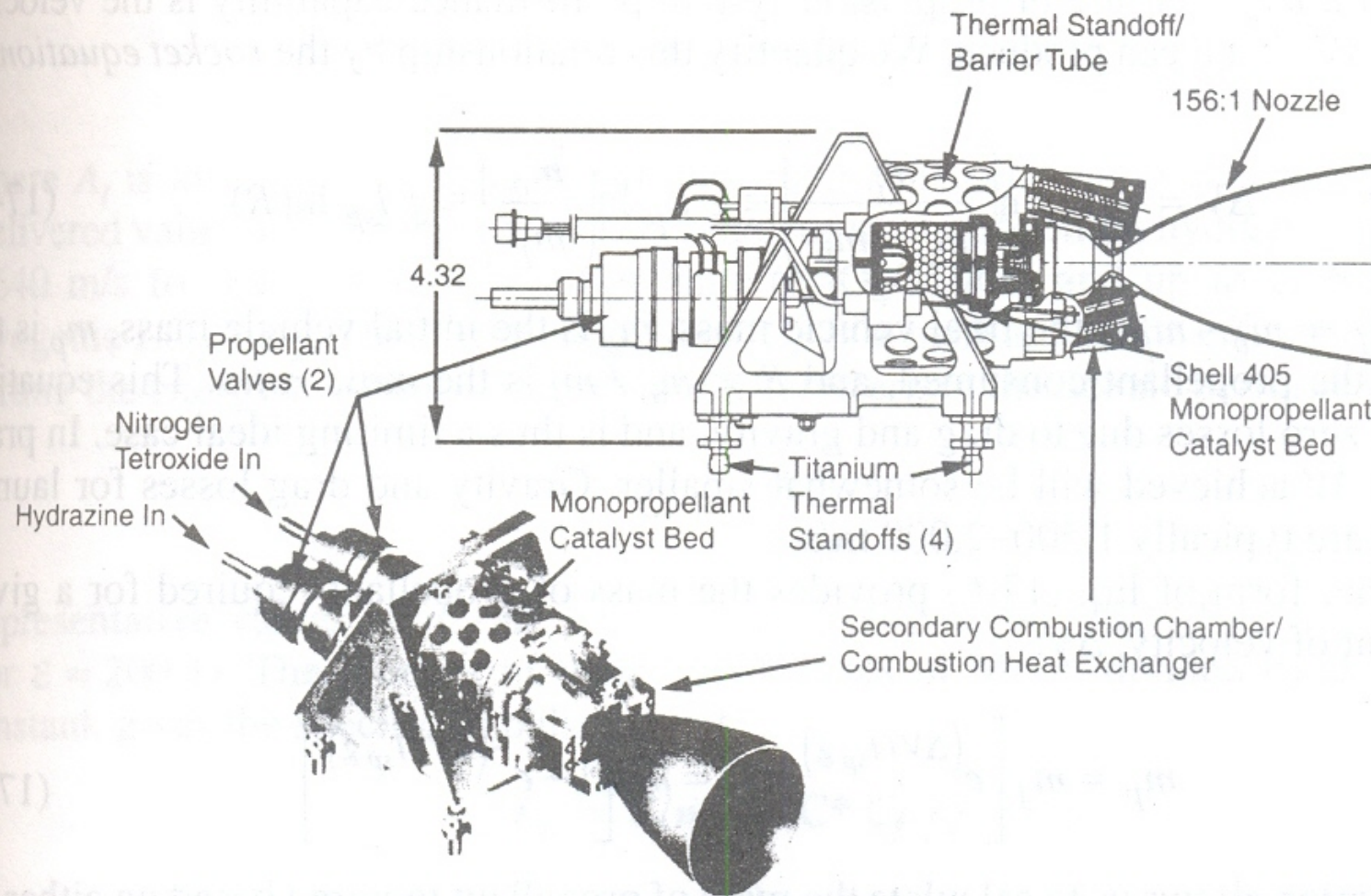
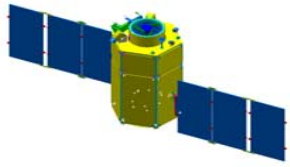


Fig. 17-1. **Secondary Combustion Augmented Thruster (SCAT).** Bimodal spacecraft attitude and velocity control RCS thruster. (Dual Mode)



17.2 Basics of Rocket Propulsion

$$\Delta V = g I_{sp} \ln \left(\frac{m_0}{m_f} \right) \quad m_p = m_0 \left[1 - e^{-(\Delta V / I_{sp} g)} \right]$$

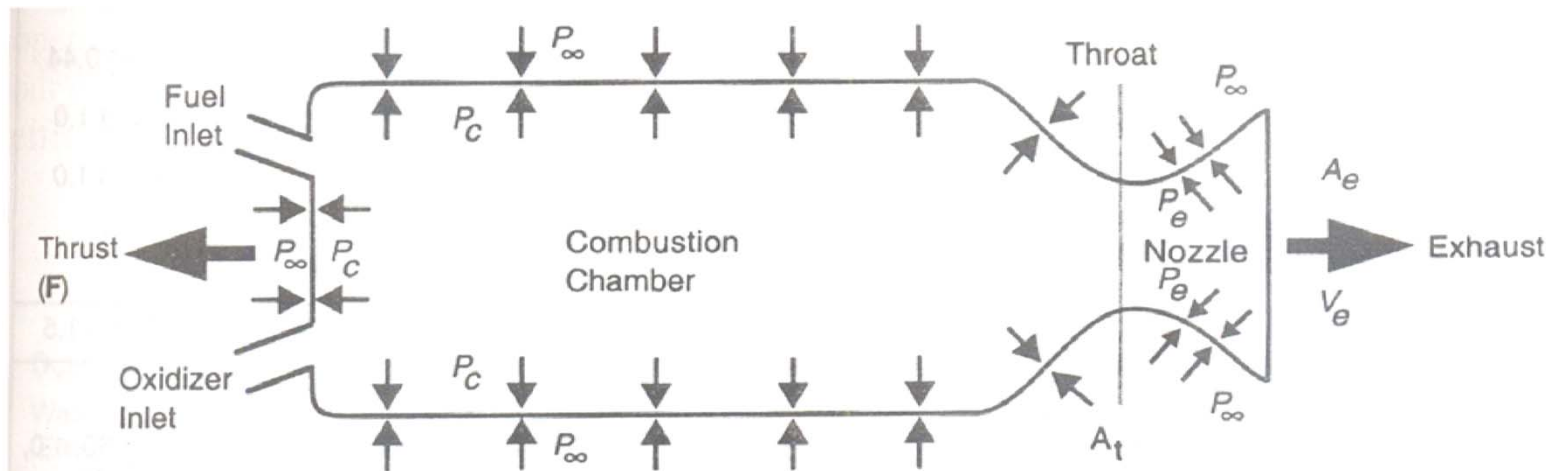
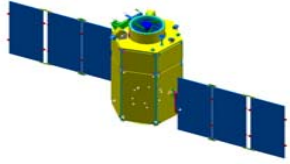
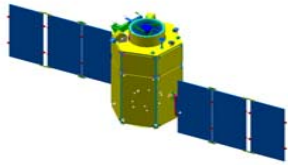


Fig. 17-2. Simplified Diagram of a Chemical Rocket. Combustion of fuel and oxidizer in the combustion chamber produces expansion of the reaction product gases which are then expelled through the nozzle. The differential between pressure inside the chamber (P_c) and at the nozzle exit (P_e) produces a reaction that propels the chamber and the vehicle in a direction opposite to that of the exhaust gases. Atmospheric pressure (P_∞) reduces the effective thrust (F), so that a rocket engine actually works more efficiently in outer space than in the Earth's atmosphere. This simplified combustion chamber diagram is also representative of a monopropellant rocket (one fuel inlet only and the upper portion of the chamber is packed with catalyst), and of a solid rocket (where there are no inlets because both the fuel and oxidizer are preloaded in the chamber and burn only upon command for ignition).



17.3 Types of Rockets

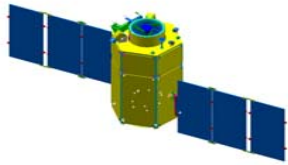
- ◆ Currently available space propulsion can be categorized as either:
 - Cold Gas Propulsion
 - Simplest form of engine
 - Just a controlled, pressurized gas source and a nozzle
 - Chemical
 - Most common systems for space applications
 - Liquid: used by engines or thrusters
 - Monopropellant/Bipropellant/Dual Mode
 - Solid: used by motors
 - Hybrid: combines advantages of Solid and Liquid
 - Electric



17.3 Types of Rockets

TABLE 17-4. Performance and Operating Characteristics of Propellants and Energy Sources for Spacecraft Propulsion Systems. These propellant and energy sources start with the low-efficiency cold gas system and work up to the highest efficiency, space-qualified system, electromagnetic.

Type	Propellant	Energy	Vacuum I_{sp} (sec)	Thrust Range (N)	Thrust Range (lb _f)	Avg Bulk Density (g/cm ³)
<i>Cold Gas</i>	N ₂ , NH ₃ , Freon, helium	High pressure	50–75	0.05–200	0.01–50	0.28*, 0.60, 0.96*
<i>Solid Motor</i>	†	Chemical	280–300	50–5 × 10 ⁶	10–10 ⁶	1.80
<i>Liquid:</i>						
Monopropellant	H ₂ O ₂ , N ₂ H ₄	Exothermic decomposition	150–225	0.05–0.5	0.01–0.1	1.44, 1.0
Bipropellant	O ₂ and RP-1	Chemical	350	5–5 × 10 ⁶	1–10 ⁶	1.14 and 0.80
	O ₂ and H ₂	Chemical	450	5–5 × 10 ⁶	1–10 ⁶	1.14 and 0.07
	N ₂ O ₄ and MMH (N ₂ H ₄ , UDMH)	Chemical	300–340	5–5 × 10 ⁶	1–10 ⁶	1.43 and 0.86 (1.0, 0.79)
	F ₂ and N ₂ H ₄	Chemical	425	5–5 × 10 ⁶	1–10 ⁶	1.5 and 1.0
	OF ₂ and B ₂ H ₆	Chemical	430	5–5 × 10 ⁶	1–10 ⁶	1.5 and 0.44
	ClF ₅ and N ₂ H ₄	Chemical	350	5–5 × 10 ⁶	1–10 ⁶	1.9 and 1.0
Dual Mode	N ₂ O ₄ /N ₂ H ₄	Chemical	330	3–200	—	1.9 and 1.0
Water Electrolysis	H ₂ O → H ₂ + O ₂	Electric / chemical	340–380	50–500	10–100	1.0



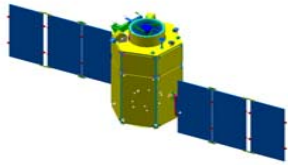
17.3 Types of Rockets

<i>Hybrid</i>	O ₂ and rubber	Chemical	225	225–3.5 × 10 ⁵	50–75,000	1.14 and 1.5
<i>Electrothermal:</i>						
Resistojet	N ₂ , NH ₃ , N ₂ H ₄ , H ₂	Resistive heating	150–700	0.005–0.5	0.001–0.1	0.28*, 0.60, 1.0, 0.019*
Arcjet	NH ₃ , N ₂ H ₄ , H ₂	Electric arc heating	450–1,500	0.05–5	0.01–1	0.60, 1.0, 0.019*
<i>Electrostatic:</i>						
Ion	Hg/A/Xe/Cs	Electrostatic	2,000–6,000	5 × 10 ^{–6} –0.5	10 ^{–6} –0.1	13.5/0.44*/2.73*/1.87
Colloid	Glycerine	Electrostatic	1,200	5 × 10 ^{–6} –0.05	10 ^{–6} –0.01	1.26
Hall Effect Thruster	Xenon	Electrostatic	1,500–2,500	5 × 10 ^{–6} –0.1	10 ^{–6} –0.02	0.22
<i>Electromagnetic:</i>						
MPD†	Argon	Magnetic	2,000	25–200	5–50	0.44*
Pulsed Plasma	Teflon	Magnetic	1,500	5 × 10 ^{–6} –0.005	10 ^{–6} –0.001	2.2
Pulsed Inductive	Argon N ₂ H ₄	Magnetic Magnetic	4,000 2,500	2–200 2–200	0.5–50 0.5–50	0.44 1.0

* Gas densities at standard conditions of pressure and temperature

† Several types in use: Organic polymers + ammonium perchlorate + powdered aluminum.

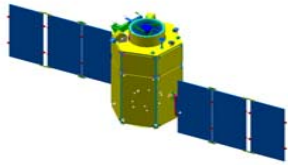
‡ MPD ≡ magnetoplasmadynamic



17.3 Types of Rockets

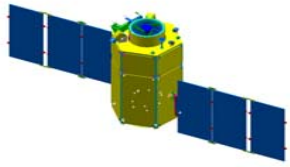
TABLE 17-5. Advantages and Disadvantages of Propellants and Energy Sources for Spacecraft Propulsion Systems.

Type	Propellant	Advantages	Disadvantages
<i>Cold Gas</i>	N ₂ , NH ₃ , Freon, helium	Extremely simple, reliable, very low cost	Very low performance, heaviest of all systems for given performance level
<i>Solid Motor</i>		Simple, reliable, relatively low cost	Limited performance, higher thrust, safety issues; performance not adjustable
<i>Liquid:</i>			
Monopropellant	H ₂ O ₂ , N ₂ H ₄	Simple, reliable, low-cost	Low performance, higher weight than bipropellant
Bipropellant	O ₂ and RP-1	High performance	More complicated system
	O ₂ and H ₂	Very high performance	Cryogenic, complicated
	N ₂ O ₄ and MMH (N ₂ H ₄ , UDMH)	Storable, good performance	Complicated
	F ₂ and N ₂ H ₄	Very high performance	Toxic, dangerous, complicated
	OF ₂ and B ₂ H ₆	Very high performance	Toxic, dangerous, complicated
	ClF ₅ and N ₂ H ₄	High performance	Toxic, dangerous
Dual Mode	N ₂ O ₄ /N ₂ H ₄	High performance	Toxic, dangerous
Water Electrolysis	H ₂ O → H ₂ + O ₂	High performance	Complicated, not developed, high power



17.3 Types of Rockets

Hybrid	O ₂ and rubber	Throttleable, nonexplosive; nontoxic, restartable	Requires oxidizer fuel system; bulkier than solids
Electrothermal:			
Resistojet	N ₂ , NH ₃ , N ₂ H ₄ , H ₂	High performance, low power, simple feed system	More complicated interfaces, more power than chemical; low thrust
Arcjet	NH ₃ , N ₂ H ₄ , H ₂	High performance, simple feed system	High power, complicated interfaces (especially thermal)
Electrostatic:			
Ion	Hg/A/Xe/Cs	Very high performance	Very high power, low thrust, complicated, not well developed
Colloid	Glycerine	Moderately high performance	High development risk, high power, complicated
Hall Effect Thruster	Xenon	High performance, relatively high power/thrust density	High development risk, high power, complicated
Electromagnetic:			
MPD	Argon	Very high performance	Very high power, high development risk, expensive, complicated
Pulsed Plasma	Teflon	High performance	Low thrust, high power, contamination, complicated
Pulsed Inductive	N ₂ H ₄ Argon	Very high performance, moderate thrust	High develop. risk, complicated, expensive, very high power

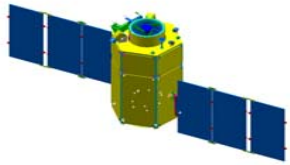


17.3 Types of Rockets

◆ (Chemical) Liquid Rockets

- Propellants are stored as liquids in tanks
- Fed on demand into the combustion chamber by gas pressurization or pump

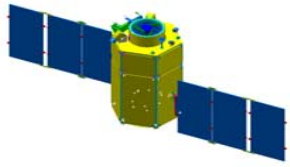
- Bipropellant Engines
 - Complex & Costly
 - Chemically react (liquid) fuel and (liquid) oxidizer
- Monopropellant Engines
 - (Solid) catalysts decompose a single (liquid) propellant
 - Hydrazine is common propellant



17.3 Types of Rockets

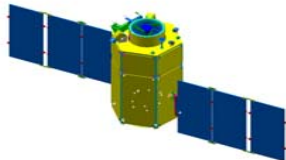
TABLE 17-6. Examples of Available Liquid Rocket Engines. For up-to-date and more detailed information, contact the developer.

Engine	Developer	Nominal Thrust (N)	Spec. Impulse (sec)	Propellants	Oper. Life (sec)	Engine Mass (kg)	Status
XLR-132	Rocketdyne	1.67×10^4	340	N_2O_4/MMH	5,000	51.26	In development
Transtar	Aerojet	1.67×10^4	330–338	N_2O_4/MMH	5,400	57.15	In development
Transtage	Aerojet	3.56×10^4	315	$N_2O_4/A-50$	1,000	107.95	Flown
Delta-II	Aerojet	4.36×10^4	320	N_2O_4/MMH	1,200	99.79	Flown
R-4D	Marquardt	4.00×10^3	309	N_2O_4/MMH	25,000	7.26	Qualified
OME/UR	Aerojet	2.67×10^4	340	N_2O_4/MMH	1,200	90.72	Modified Orbiter maneuvering engine



17.3 Types of Rockets

RL10-A	Pratt & Whitney	7.34×10^4	446	LO ₂ /LH ₂	400	138.35	Flight qualified (Centaur)
DM/LAE	TRW	4.45×10^2	315	N ₂ O ₄ /N ₄ H ₄	15,000	4.54	Flown
R4-D	Marquardt	4.89×10^2	310	N ₂ O ₄ /MMH	20,000	3.76	Flown
R42	Marquardt	8.90×10^2	305	MON-3/MMH	15,000	4.54	Qualified
MMBPS	TRW	4.45×10^2	302	N ₂ O ₄ /MMH	20,000	5.22	Flight qualified
RS-41	Rocketdyne	1.11×10^4	312	N ₂ O ₄ /MMH	2,000	113.40	Flight qualified (Peacekeeper)
ADLAE	TRW	4.45×10^2	330	N ₂ O ₄ /N ₂ H ₄	28,000	4.50	In qual.
Chandra X-Ray Observatory	TRW	4.25×10^3	322.5	N ₂ O ₄ /N ₂ H ₄	25,000	4.5	Flight qualified
HS 601 AKE	ARC/LPG	4.89×10^2	312	N ₂ O ₄ /MMH	10,000	4.08	In development
R-40A	Marquardt	4.00×10^3	309	N ₂ O ₄ /MMH	25,000	7.26	Qualified (mod. of Shuttle RCS engine)
HPLAM	TRW	4.45×10^2	325	N ₂ O ₄ /MMH	30,000	4.60	In advanced development



17.3 Types of Rockets

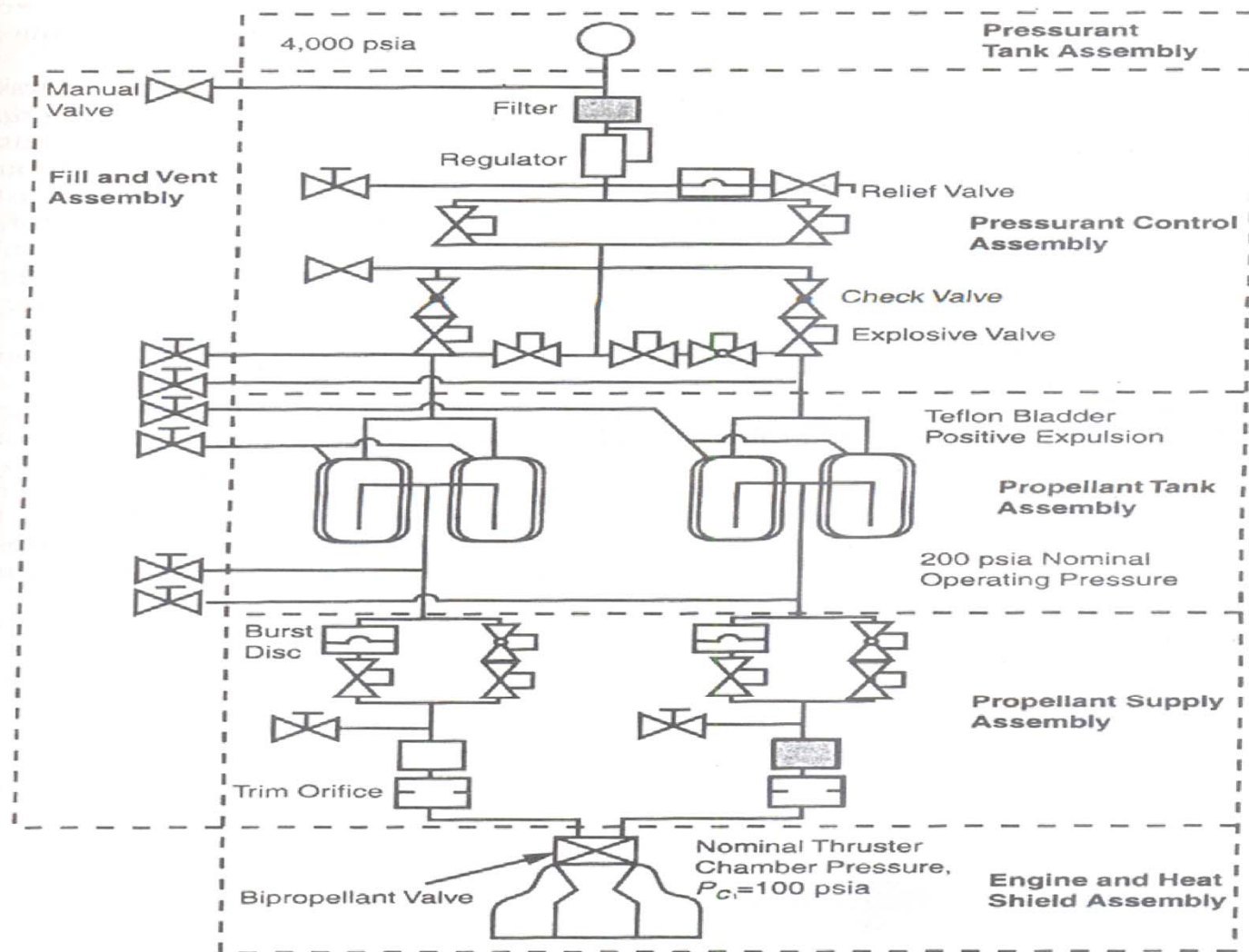
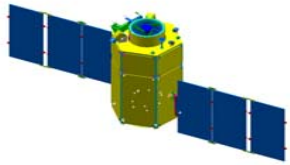


Fig. 17-3. Pressure-Fed Propulsion System Using Earth-Storable **Bipropellant** (N_2O_4/MMH). Propulsion system designers trade-off reliability and safety with complexity and mass.



17.3 Types of Rockets

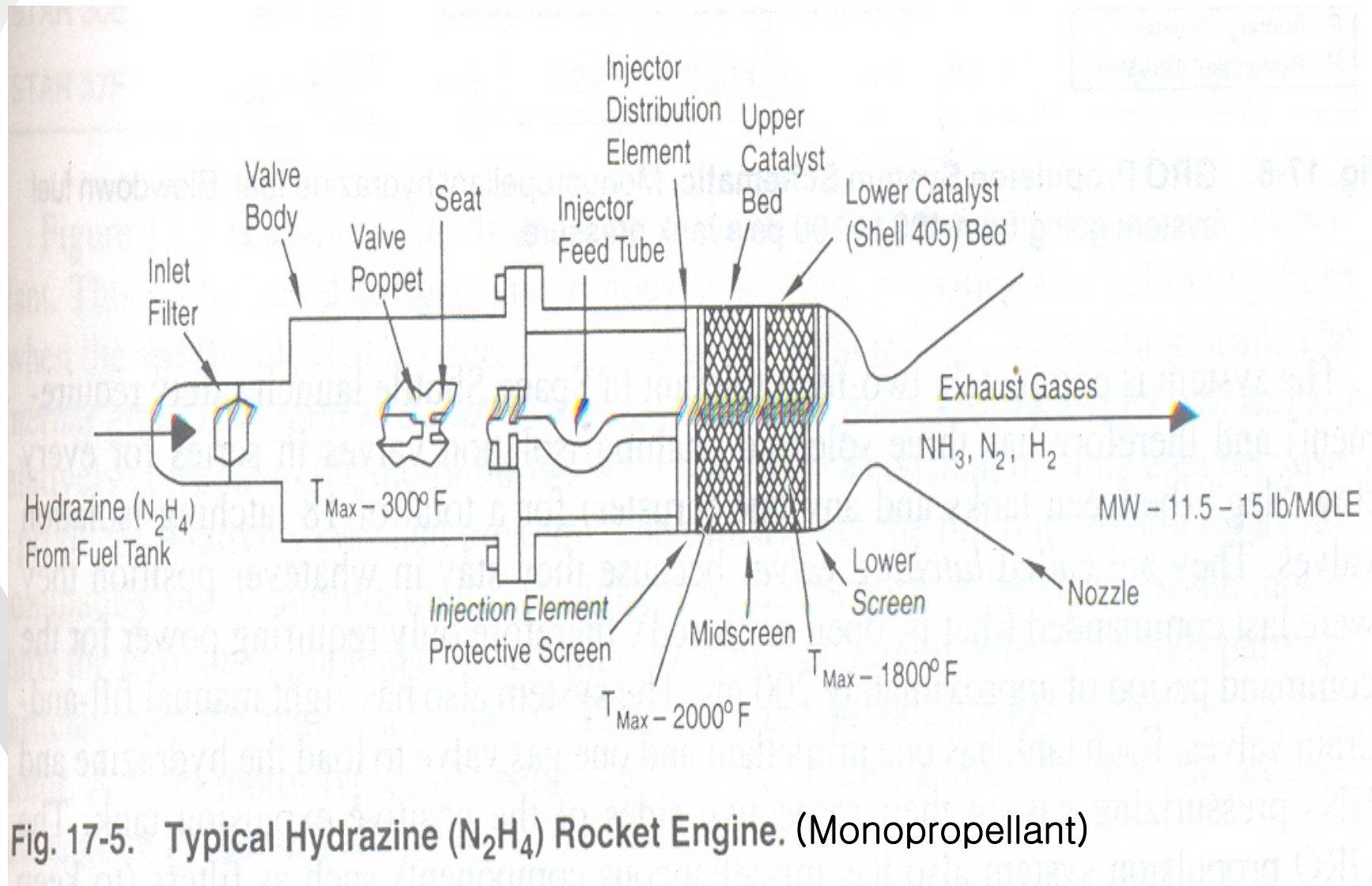
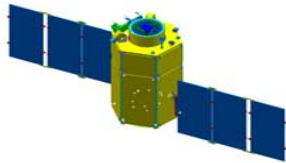


Fig. 17-5. Typical Hydrazine (N_2H_4) Rocket Engine. (Monopropellant)



17.3 Types of Rockets

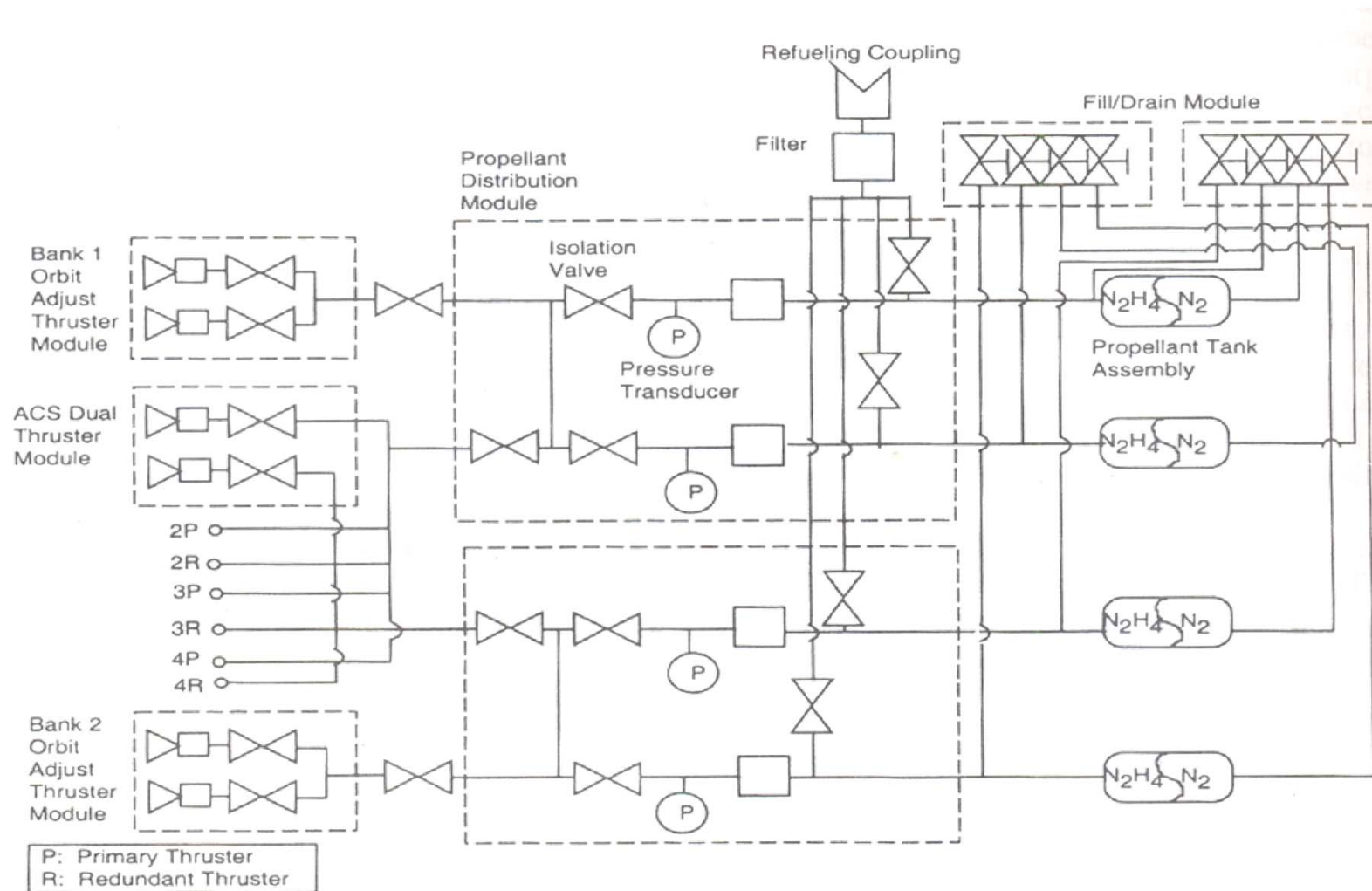
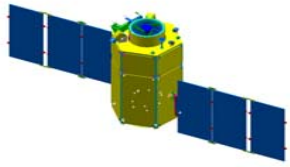


Fig. 17-6. GRO Propulsion System Schematic. Monopropellant hydrazine fuel. Blowdown fuel system going from 400 to 100 psia tank pressure.



17.3 Types of Rockets

- ◆ (Chemical) Solid Rockets
 - Store propellants in solid form
 - Grain composed of fuel and oxidizer
 - Powdered Aluminum is a typical fuel
 - Ammonium Perchlorate is an oxidizer

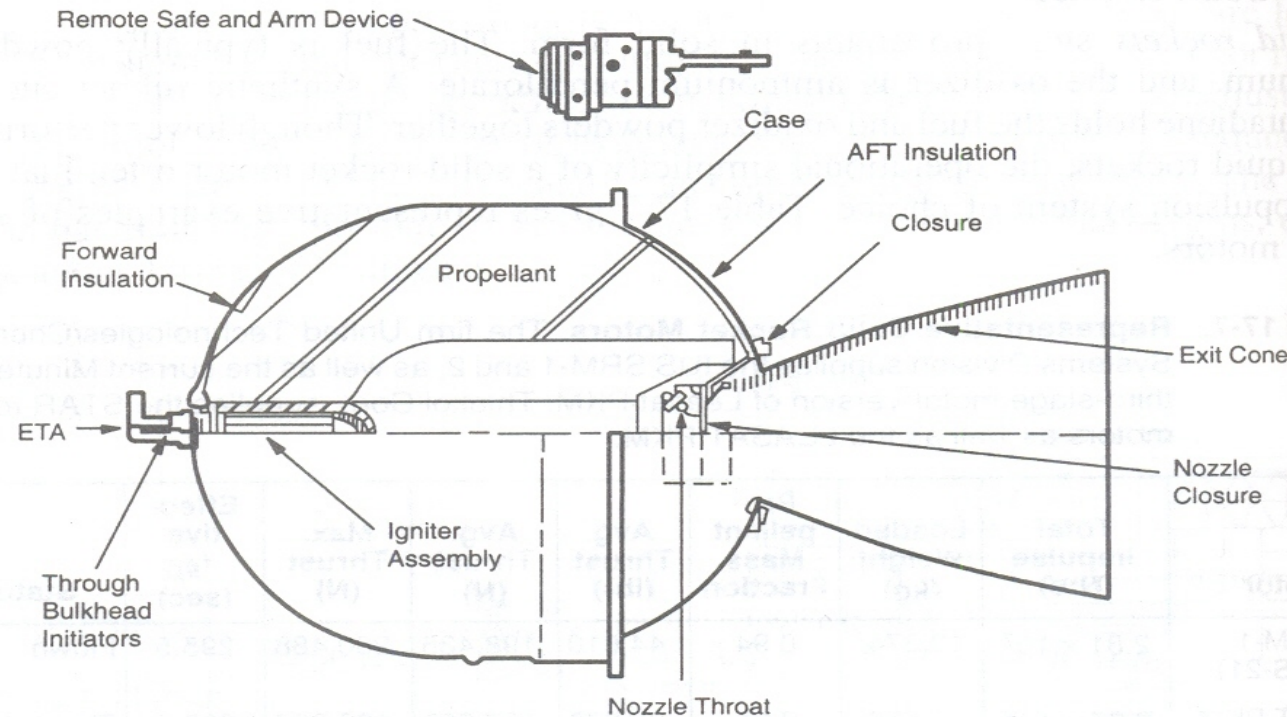
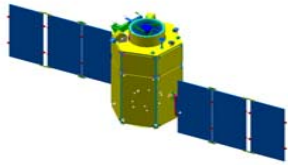


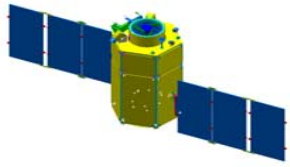
Fig. 17-7. Typical Solid Propellant Rocket Motor. ETA is the Explosive Transfer Assembly.



17.3 Types of Rockets

TABLE 17-7. Representative Solid Rocket Motors. The firm United Technologies/Chemical Systems Division supplies the IUS SRM-1 and 2, as well as the current MinuteMan third-stage motor version of Leasat PKM. Thiokol Corp. supplies the STAR rocket motors as well as the LEASAT PKM.

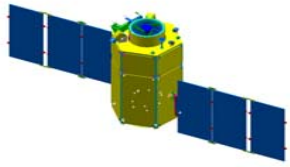
Motor	Total Impulse (N·s)	Loaded Weight (kg)	Pro-pellant Mass Fraction	Avg. Thrust (lbf)	Avg. Thrust (N)	Max. Thrust (N)	Effective I_{sp} (sec)	Status
IUS SRM-1 (ORBUS-21)	2.81×10^7	10,374	0.94	44,610	198,435	260,488	295.5	Flown
LEASAT PKM	9.26×10^6	3,658	0.91	35,375	157,356	193,200	285.4	Flown
STAR 48A	6.78×10^6	2,559	0.95	17,900	79,623	100,085	283.9	Flown
STAR 48B(S)	5.67×10^6	2,135	0.95	14,845	66,034	70,504	286.2	Qualified
STAR 48B(L)	5.79×10^6	2,141	0.95	15,160	67,435	72,017	292.2	Qualified
STAR 62	7.12×10^6	2,459					293.5	In develop.
STAR 75	2.13×10^7	8,066	0.93	44,608	198,426	242,846	288.0	In develop.
IUS SRM-2 (ORBUS-6)	8.11×10^6	2,995	0.91	18,020	80,157	111,072	303.8	Flown
STAR 13B	1.16×10^5	47	0.88	1,577	7,015	9,608	285.7	Flown
STAR 30BP	1.46×10^6	543	0.94	5,960	26,511	32,027	292.0	Flown
STAR 30C	1.65×10^6	626	0.95	7,140	31,760	37,031	284.6	Flown
STAR 30E	1.78×10^6	667	0.94	7,910	35,185	40,990	289.2	Flown
STAR 37F	3.02×10^6	1,149	0.94	9,911	44,086	49,153	291.0	Flown



17.3 Types of Rockets

◆ (Chemical) Hybrid Rockets

- Propellants are stored in different forms; Normally
 - Fuel is solid
 - Oxidizer is a liquid or gas
- Advantages
 - Safety
 - Throttling
 - Restart
 - Storability
 - Environmentally Clean



17.3 Types of Rockets

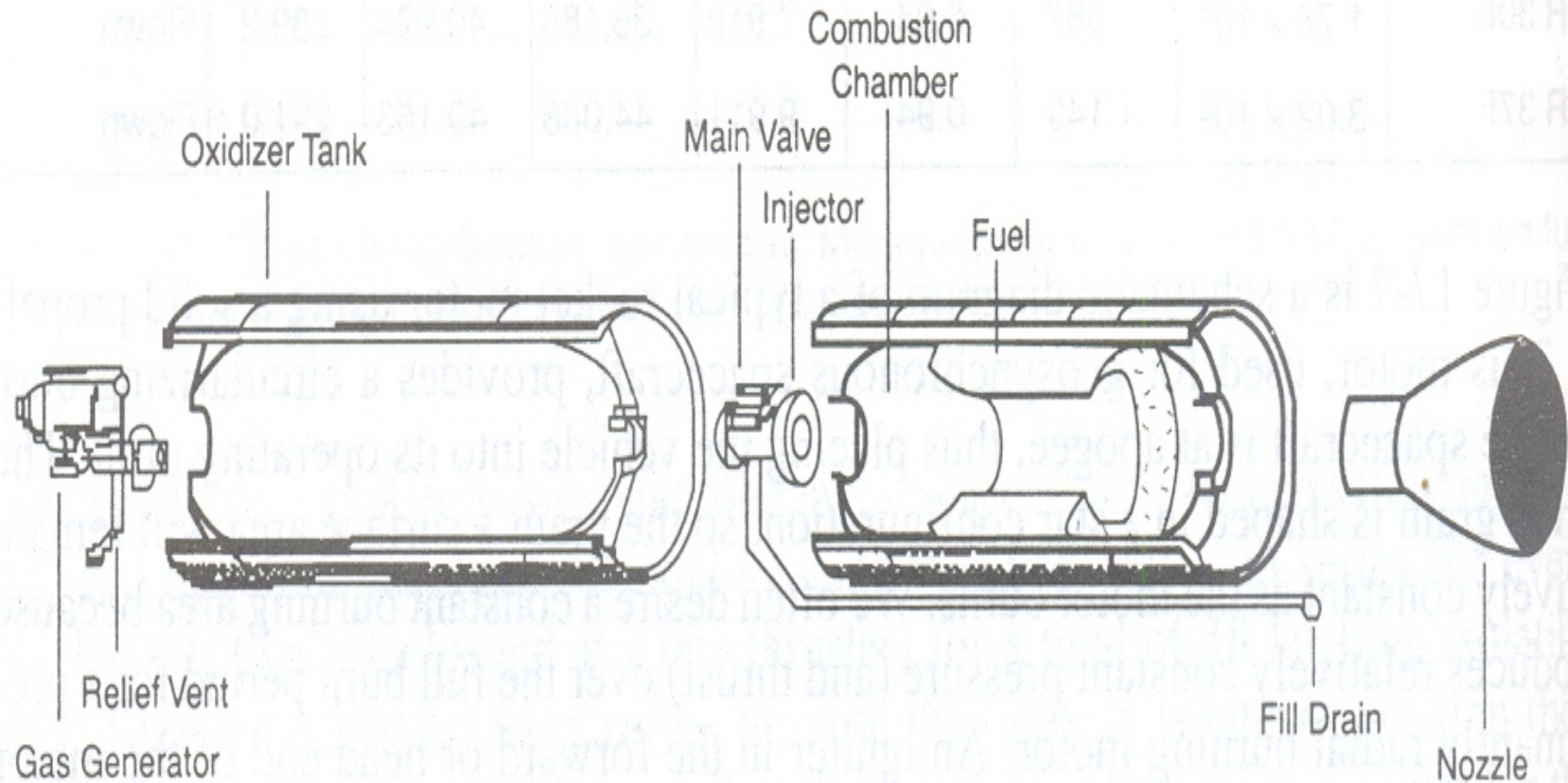
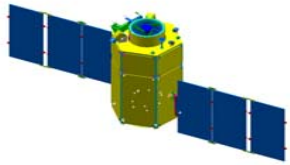


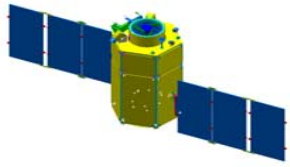
Fig. 17-8. Schematic Drawing of a Hybrid Rocket.



17.3 Types of Rockets

TABLE 17-8. Representative Hybrid Rockets.

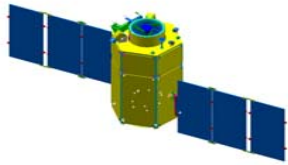
Motor	Average Thrust (lb _f)	Average Thrust (kN)	Burn Duration (sec)	Fuel	Oxidizer	Comments
<i>American Rocket Company</i>						
H-500	75,000	333	70	HTPB	LOx	Qualified for flight
H-250	32,000	142		HTPB	LOx	In development
H-50	10,000	44		HTPB	LOx	In development
U-50	6,500	29		HTPB	LOx	In development
U-1	100	0.44		HTPB	LOx	In development
<i>United Technologies</i>						
	40,000	178	300	HTPB	IRFNA	Flown on Firebolt air-launched target drone, 1968
<i>StarsTruck</i>						
	40,000	178		CTBN	LOx	Flown on Dolphin water-launched sounding rocket, 1984
<i>USAF Academy</i>						
H-1	55	0.25	2.3	HTPB	GOx	Flown on 4-ft tall rocket for student project, 1991



17.3 Types of Rockets

◆ Electric Propulsion

- Uses externally provided electrical power either from:
 - Sun
 - Nuclear or Thermodynamic Conversion Thermal Engines
- To accelerate the working fluid to produce useful thrust
- Ion Engine
 - Electric field accelerates charged particles which exit at high velocity
- MPD (Magneto-Plasma-Dynamics) Thruster
 - Current-carrying plasma interacts with a magnetic field resulting in a Lorentz acceleration to expel the plasma
- No fundamental limit to the exhaust velocity that we can obtain with an electric rocket
- However, the power required may grow to the point where further acceleration is pointless
 - Optimum Exhaust Velocity (Optimum Specific Impulse) depends on the power



17.3 Types of Rockets

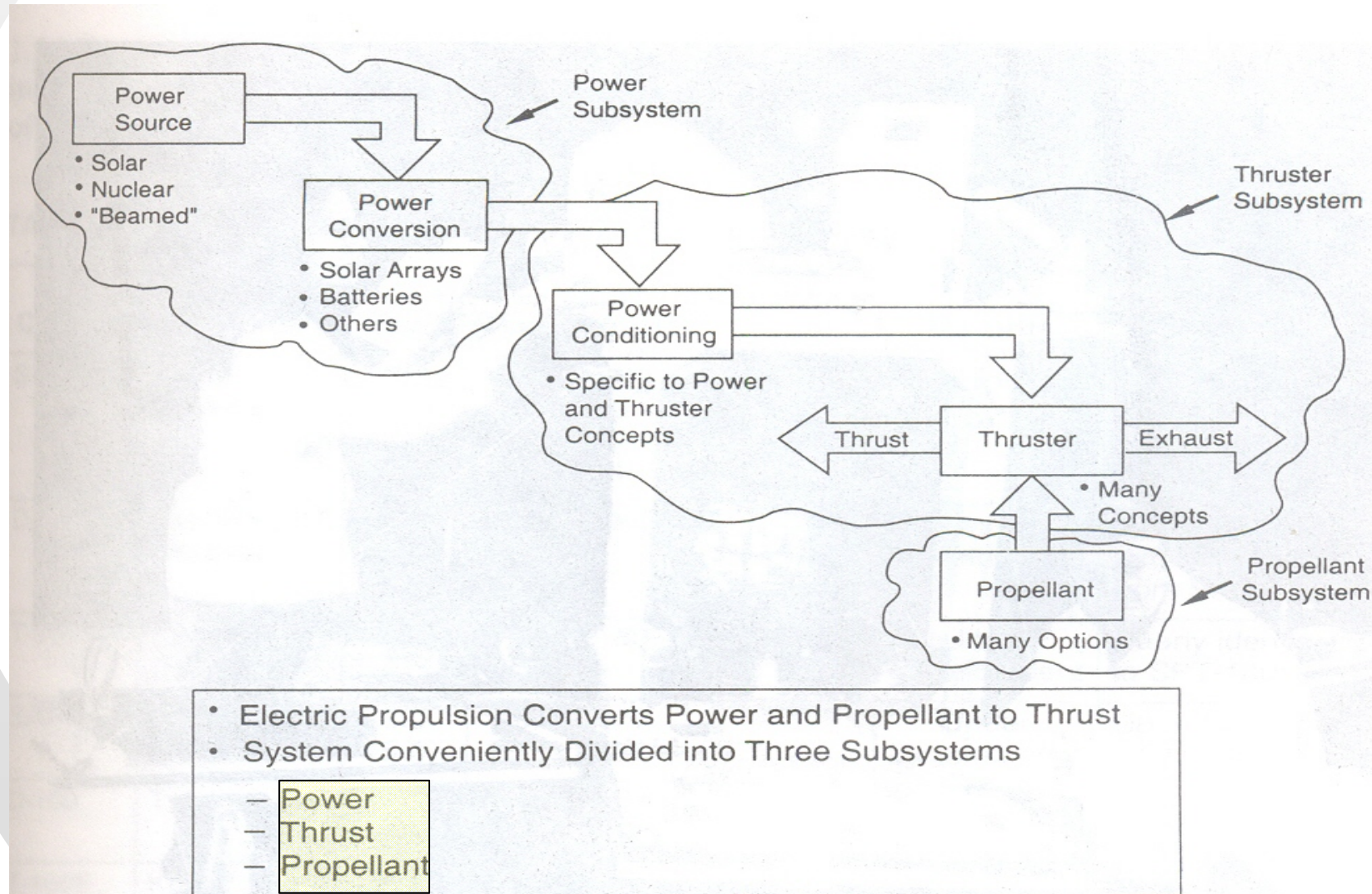
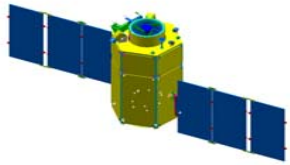


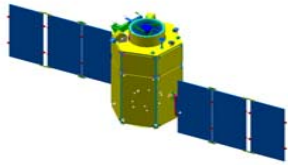
Fig. 17-10. **Generic Electric Propulsion System.** Functional Block Diagram.



17.3 Types of Rockets

TABLE 17-9. Electric Propulsion; Three Classes of Accelerator Concepts.

	Electrothermal	Electrostatic	Electromagnetic
<i>Mechanism</i>	• Gas heated via resistance element or arc and expanded through nozzle • Resistojets • Arcjets	• Ions electrostatically accelerated • Hall effect (HET) • Ion • Field emission	• Plasma accelerated via interaction of current and magnetic field • Pulsed plasma (PPTs) • Magnetoplasmadynamic (MPD) • Pulsed inductive (PIT)
<i>Power</i>	0.4–2 kW	1–50 kW	50 kW–1 MW
<i>Specific Impulse, I_{sp}</i>	300–800 sec	1,000–3,000 sec	2,000–5,000 sec



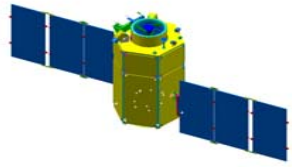
17.3 Types of Rockets

TABLE 17-10. Characteristics of Selected Electric Propulsion Flight Systems.

Concept	Characteristics					
	Specific Impulse, (sec)	Input Power, (kW)	Thrust/Power, (mN/kW)	Specific Mass, (kg/kW)	Propellant	Supplier
<i>Resistojet</i>	296	0.5	743	1.6	N ₂ H ₄	Primex
	299	0.9	905	1	N ₂ H ₄	Primex, TRW
<i>Arcjet</i>	480	0.85	135	3.5	NH ₃	IRS/ITT
	502	1.8	138	3.1	N ₂ H ₄	Primex
	>580	2.17	113	2.5	N ₂ H ₄	Primex
	800	26*	—	—	NH ₃	TRW, Primex, CTA
<i>Pulsed Plasma Thruster (PPT)</i>	847	< 0.03†	20.8	195	Teflon	JHU/APL
	1,200	< 0.02†	16.1	85	Teflon	Primex, TSNIIMASH, NASA
<i>Hall Effect Thruster (HET)</i>	1,600	1.5	55	7	Xenon	IST, Loral, Fakel
	1,638	1.4*	—	—	Xenon	TSNIIMASH, NASA
	2,042	4.5	54.3	6	Xenon	SPI, KeRC
<i>Ion Thruster (IT)</i>	2,585	0.5	35.6	23.6	Xenon	HAC
	2,906	0.74	37.3	22	Xenon	MELCO, Toshiba
	3,250	0.6	30	25	Xenon	MMS
	3,280	2.5	41	9.1	Xenon	HAC, NASA
	3,400	0.6	25.6	23.7	Xenon	DASA

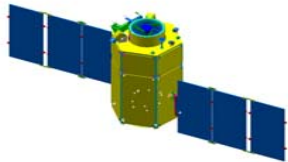
* Thruster input power.

† Power dependent on pulse rate.



17.4 Component Selection & Sizing

- ◆ Simplest way to feed the propellant to the thrust chamber is to displace it from its storage vessels with a high-pressure gas
- ◆ Alternative systems employs a pump (a piston or turbo-pump) driven by:
 - Turbines
 - Obtain drive power either from:
 - Hot-Gas Cycle (most common)
 - Electromechanical Actuators (receive power from batteries or solar array)
 - Gas Pressure Intensifiers
 - Electric motors



17.4 CS&S

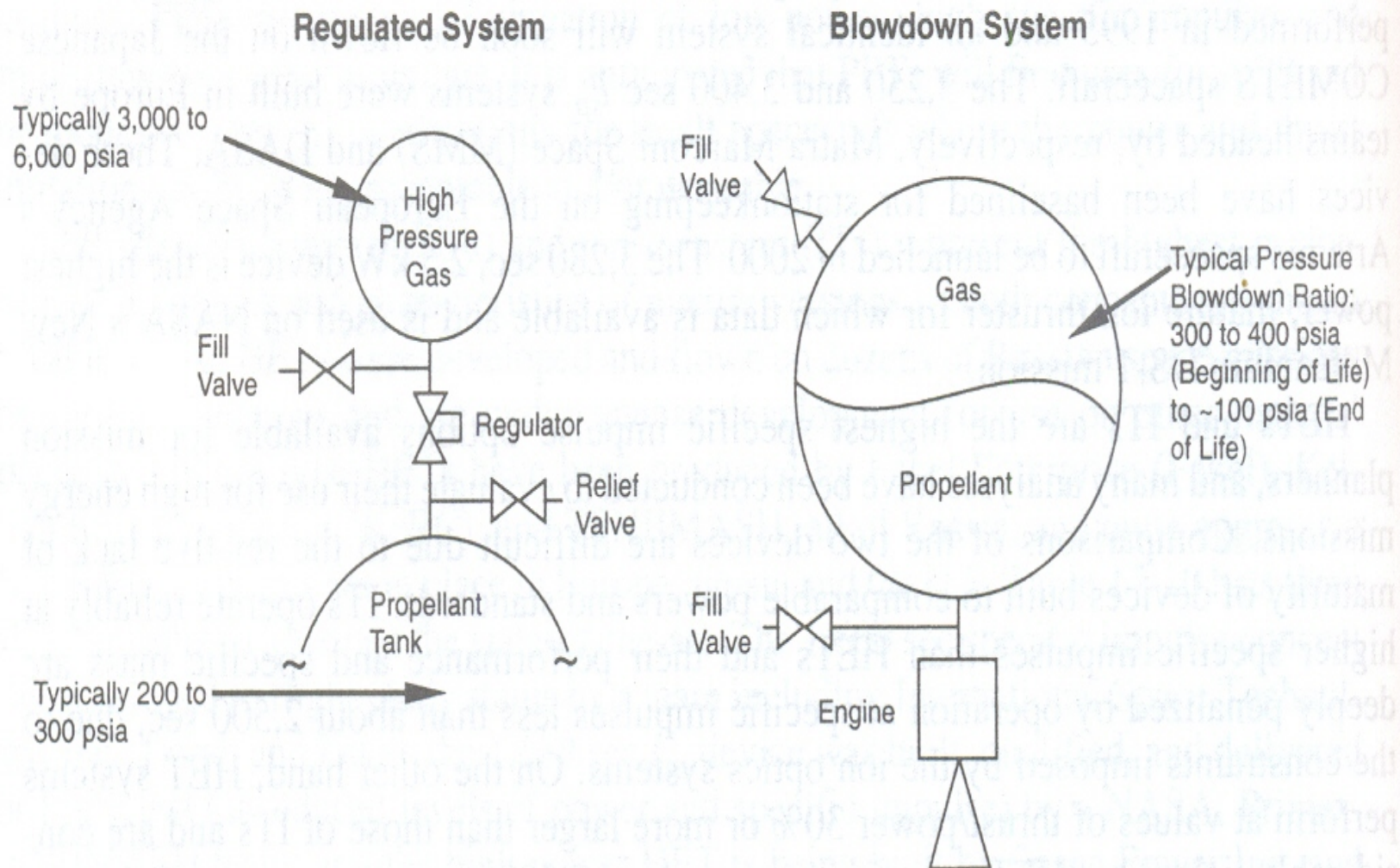
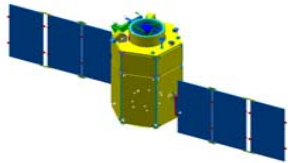


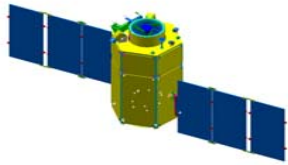
Fig. 17-12. Pressurization Systems.



17.4 CS&S

TABLE 17-12. Available Options for Positive Expulsion Tanks. ΔP = pressure difference. After deciding that our spacecraft requires a liquid propulsion system, we must decide on system characteristics.

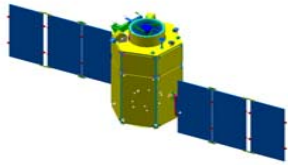
Tank Option	Advantages	Disadvantages	Typical Applications
<i>Metal Diaphragm</i>	• High volume efficiency • Good center of gravity control • No ullage volume • No sliding seals • Proven design	• High weight • High cost • High-expulsion ΔP • Optimizes only for special envelope	• Spacecraft control & maneuvering • Launch vehicles • Upper stages • Missiles
<i>Rolling Diaphragm</i>	• Light weight • Low cost • Low ΔP during expulsion	• Inspection of internal welds	• Missile interceptors • Maneuvering missiles
<i>Piston</i>	• Extensive database • Low ΔP during expulsion • Design adapts easily to growth	• High cost • Low volumetric efficiency • Critical tolerance on shell • Sliding seals possible blowby	• High acceleration missiles
<i>Rubber Diaphragm</i>	• Extensive database • Low ΔP during expulsion • Not cycle limited • Proven design • High expulsion efficiency	• Compatibility limits on propellants	• Spacecraft control & maneuvering • Launch vehicles • Upper stages
<i>Metal Bellows</i>	• No sliding seals • Good center of gravity control • Proven design • Good compatibility • Hermetically sealed	• High weight • High cost • Limited cycle capability • Low volumetric efficiency	• Missiles • Spacecraft • Launch vehicles



17.4 CS&S

TABLE 17-13. Flight-Qualified Thrusters Used to Control Spacecraft Attitude and Velocity. Thruster life is expressed in terms of total impulse or number of pulses.

Nom. Thrust (N)	Nom. Thrust (lb _f)	Propellants*	Developers	Total Impulse (10 ³ N·s)	<i>I</i> _{sp} Range (sec)		Weight (kg)
					Steady State	Pulse†	
0.09–0.22	0.02–0.05	Mono H	TRW, Olin/RRC ERNO, Marquardt, HAC	4–200	205–215	110–180	0.1–0.2
0.09–0.67	0.02–0.150	Mono H‡	TRW, Olin/RRC	90–800	285–320	250–290	0.5–0.9
2.22	0.5	Mono H	Olin/RRC, Marquardt	40–200	215–230	120–200	0.1–0.2
4.45	1.0	Mono H	TRW, SEP, HAC, Marquardt, Olin/RRC	40–1,100	210–230	120–210	0.1–0.2
13–18	3–4	Mono H	TRW, SEP, ERNO	40–1,100	215–235	150–210	0.2–0.3
22–36	5–8	Mono H	TRW, Olin/RRC, Marquardt, HAC	40–1,100	215–240	120–210	0.2–0.3
16–50	4–11	Bimodal	TRW (SCAT)	2.3 × 10 ⁶	305–325	310	2
45–67	10–15	Mono H	Marquardt, TRW	40–1,300	215–240	120–210	0.3–0.5
111	25	Mono H	Olin/RRC	40–400	215–240	150–220	1.5–1.6
133	30	Mono H	Marquardt	40–300	225–242	150–225	2–3
178–222	40–50	Mono H	Olin/RRC, TRW	40–200	220–245	150–220	1.4–1.8



17.4 CS&S

178–222	40–50	Mono H	Olin/RRC, TRW	40–200	220–245	150–220	1.4–1.8
445–689	100–155	Mono H	Olin/ RRC, Marquardt	1,100	225–245	150–225	1.8–2.3
1,335	300	Mono H	Walter Kidde, Olin/RRC	2,200	225–245	150–225	11.3
2,669	600	Mono H	Olin/RRC	200	225–240	N/A	8.2
11	2.5	N ₂ O ₄ /MMH	MMB	200	285	210	0.5
22	5.0	N ₂ O ₄ /MMH	Marquardt, ARC/LPG	200–400	290	220	0.7
111	25.0	N ₂ O ₄ /MMH	Marquardt	1,100	300	220	1.4
400–489	90–110	N ₂ O ₄ /MMH	ARC/LPG	9,000	308	N/A	4.1
450	100	N ₂ O ₄ /MMH	MBB	7,000	305	N/A	4.5
440	105	N ₂ O ₄ /MMH	TRW	11,000	325	N/A	4.5
450	110	N ₂ O ₄ /MMH	Marquardt	900	309	N/A	3.8
445	105	N ₂ O ₄ /N ₂ H ₄	TRW	10,000	322	N/A	4.5
440	105	N ₂ O ₄ /N ₂ H ₄	TRW	10,000	330	N/A	5.5
450	110	N ₂ O ₄ /N ₂ H ₄	Royal Ordnance	7,000	317	N/A	4

* *Mono H* is monopropellant hydrazine (N₂H₄) and *MMH* is monomethyl hydrazine.

† For low duty cycles (<10%) and short pulse width (10–15 ms), use the lower pulsed I_{sp} . For duty cycles >10% and pulse widths greater than 50 ms, use the higher pulsed I_{sp} .

‡ Electrothermally augmented (resistojet/ EHT)